

Lebanese University – Faculty of Engineering

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Subject: Sensors

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Sensors project report (Greenhouse)

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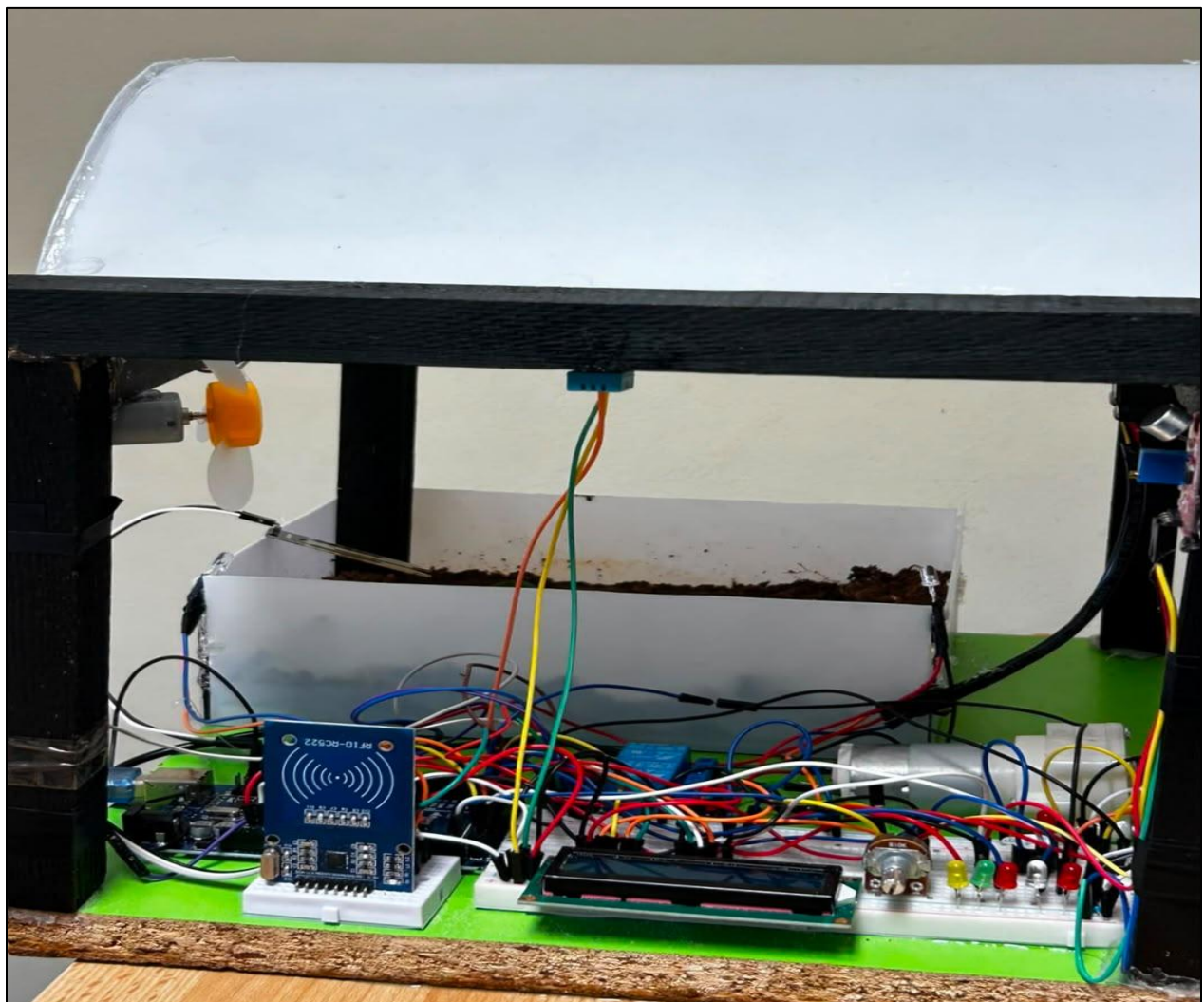
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1. INTRODUCTION:

Modern agriculture increasingly relies on precision automation to optimize crop yields and conserve resources. This project presents the design and implementation of an automated smart greenhouse system utilizing the Arduino Mega 2560 microcontroller. The system integrates an array of environmental sensors and actuators to regulate microclimate conditions, manage automated irrigation, track water consumption, and provide localized perimeter security.



2. PROBLEM STATEMENT:

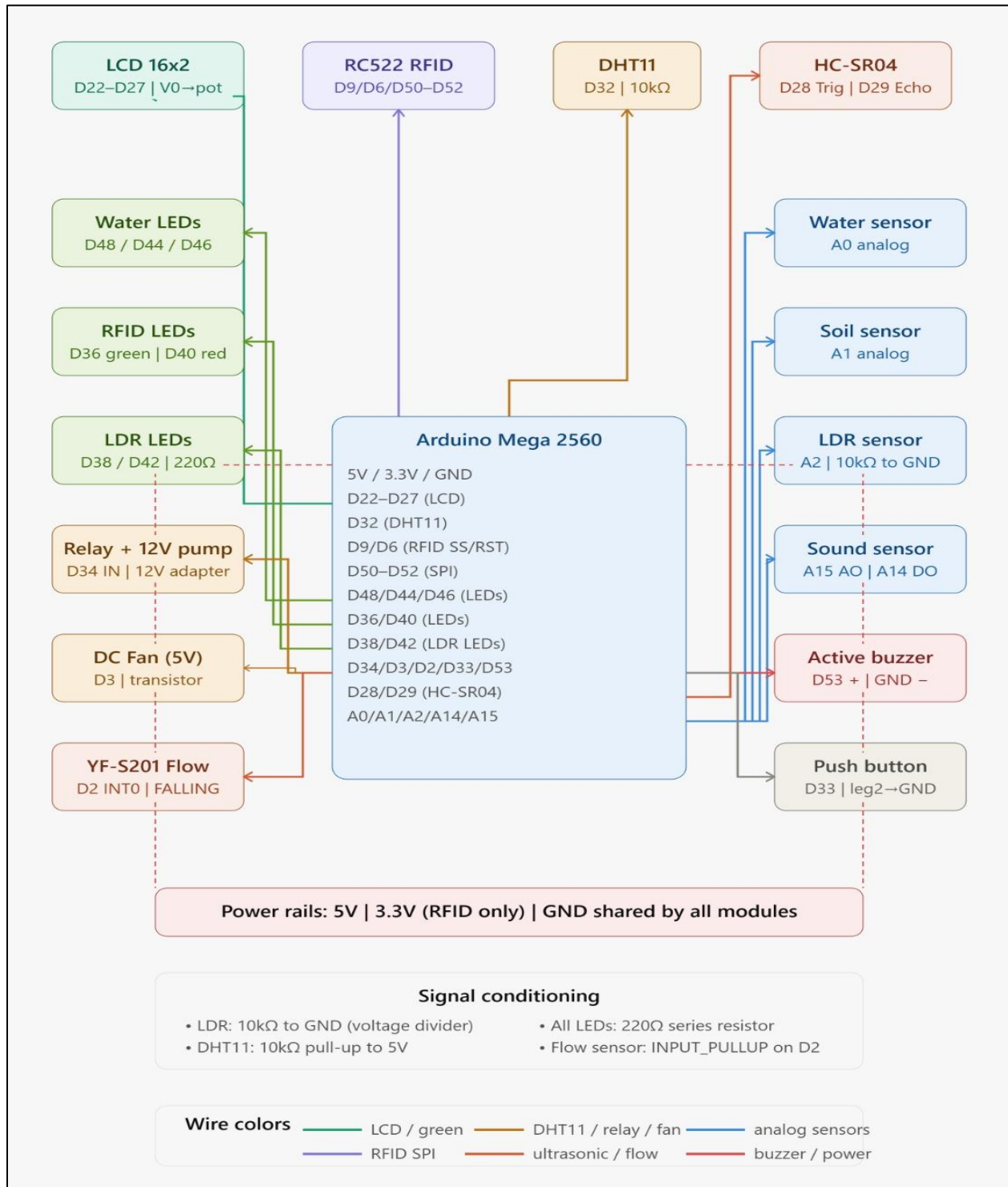
Traditional greenhouse management relies heavily on manual labor and periodic human observation, which introduces significant risks:

- **Delayed Response:** Belated detection of thermal stress or soil dehydration can severely hinder plant growth.
- **Resource Inefficiency:** Manual watering often results in over-irrigation, wasting water and leaching soil nutrients.
- **Security & Auditing Risks:** Greenhouses lack automated entry logs or environmental tamper alarms, leaving crops vulnerable to unauthorized access or pests.



The objective of this project is to engineer an autonomous closed-loop system that continuously monitors environmental metrics, dynamically regulates climate parameters, logs access controls, and ensures safe resource deployment without human intervention.

3. SYSTEM ARCHITECTURE:



The system follows a classic Sensing, Processing, Actuation control loop pattern. The Arduino Mega 2560 acts as the central

controller. Sensors connect via analog inputs (A0–A2, A15), digital I/O (D2–D53), SPI (RC522 RFID), and the hardware interrupt pin (D2 for flow sensor). A 12V external adapter powers the irrigation pump through a 2-channel relay module. A 16x2 HD44780 LCD presents six pages of data navigated by a push button. All outputs are controlled without blocking delays, using millis()-based timers for responsiveness.

Component	Arduino Pin(s)
LCD 16x2	D22,D23,D24,D25,D26,D27
DHT11	D32
RC522 RFID	D9,D6,D50-D52
Water LEDs	D48,D44,D46
RFID Green LED	D36
RFID Red LED	D40
LDR LEDs	D38,D42
Soil Relay	D34
Fan	D3
Flow Sensor	D2
HC-SR04	D28,D29
Buzzer	D53
Push Button	D33
Water Level	A0
Soil Moisture	A1
LDR	A2
Sound AO	A15
Sound DO	A14

figure showing the wiring map

4. SENSORS SELECTION:

Sensor Module	Parameter Measured	Signal Type	Engineering Justification
DHT11	Ambient Temp / Humidity	Digital (Single-Wire)	Low-cost integrated climate tracking.
Soil Sensor	Volumetric Soil Moisture	Analog (A1)	Direct corrosion-resistant capacitive/resistive tracking.
Water Sensor	Reservoir Level	Analog (A0)	Prevents pump dry-running by monitoring water availability.
LDR Sensor	Ambient Illumination	Analog (A2)	Easily mapped using a voltage divider to manage shading/LEDs.
Sound Sensor	Acoustic Thresholds	Analog/Digital	Detects physical structural anomalies or environmental noise.
HC-SR04	Target Distance / Presence	Digital (D28/D29)	Non-contact ultrasonic proximity tracking for security.
YF-S201	Water Flow Rate	Pulse / Interrupt (D2)	Provides real-time feedback on precise volume of water delivered.

5. HARDWARE DESIGN AND SCHEMATICS:

The physical hardware layout utilizes dedicated power and signal routing to avoid logic interference:

- **Power Distribution:** A shared global ground reference is maintained across all modules. The **RC522 RFID** module is strictly isolated to the 3.3V rail, while all other primary modules operate on the 5V rail.
- **Actuator Isolation:** The 12V high-current inductive water pump is isolated from the microcontroller logic pins using a mechanical relay module driven via D34, drawing power from an external independent 12V DC adapter.
- **Signal Conditioning:** The LDR uses a 10 K Ω pull-down resistor to form a stable voltage divider.
 - The DHT11 features a 10 K Ω pull-up resistor on its data line to prevent floating states.
 - All visual indicators (Water, RFID, and LDR LEDs) feature 220 Ω current-limiting series resistors to protect the microcontroller's I/O pins from overcurrent.

6. SOFTWARE DESIGN AND FIRMWARE ARCHITECTURE:

The firmware is written in Arduino C++ and structured into six functions called from the main loop:

Function	Description
checkRFID()	Reads RFID card, enables system on valid UID, shows green/red LED + LCD message
readAllSensors()	Reads all sensors, DHT11 every 2 seconds, flow every 1 second and others every loop
controlOutputs()	All actuation logic: fan, relays, LEDs, buzzer – all non-blocking
handleButton()	Detects button press, cycles LCD pages with 50ms debounce
updateLCD()	Refreshes LCD every 500ms, renders current page data
getDistance()	Sends ultrasonic pulse, measures echo duration, returns distance in cm

Key design decision: all timed actions (buzzer, fan, LED) use `millis()`-based non-blocking timers instead of `delay()`. This ensures the system remains responsive to button presses and sensor readings at all times. The flow sensor uses a hardware interrupt (`attachInterrupt` on D2, `FALLING` edge) to count pulses without missing any at high flow rates. The interrupt is briefly detached during the 1-second calculation window to prevent data corruption.

- LCD page structure:

Page	Content	Trigger
0	Temperature ($^{\circ}\text{C}$), fan status (on/off), humidity (%)	Default on startup
1	Flow rate (L/min), total volume (L)	Button press x1
2	Water tank level (empty/low/medium/full)	Button press x2
3	Soil moisture (dry/wet), pump status (on/off)	Button press x3
4	Ultrasonic distance (cm), object detection status	Button press x4
5	Sound level (raw value 0-1023), loud/quiet status	Button press x5

7. CALIBRATION PROCEDURES AND RESULTS:

Each analog sensor was calibrated using three known reference points. Raw sensor values were recorded, compared to reference measurements, and a linear calibration equation derived.

Sensor	Ref. Point 1	Ref. Point 2	Ref. Point 3	Calibration Equation
DHT11 Temp	0 $^{\circ}\text{C}$ → raw 1 $^{\circ}\text{C}$	25 $^{\circ}\text{C}$ → raw 24 $^{\circ}\text{C}$	40 $^{\circ}\text{C}$ → raw 39 $^{\circ}\text{C}$	$T_{\text{cal}} = T_{\text{raw}} + 1.0^{\circ}\text{C}$
DHT11 Hum	30% → raw 29%	55% → raw 54%	80% → raw 79%	$H_{\text{cal}} = H_{\text{raw}} + 1.0\%$
LDR (A2)	Dark: raw 12	Room: raw 450	Torch: raw 980	Threshold set at 10 (raw)
Soil (A1)	Dry: raw 680	Damp: raw 420	Wet: raw 280	Dry threshold = 500 (raw)
Water level	Empty: raw 80	Half: raw 450	Full: raw 820	Low=300, Med=600, High=700
YF-S201	0.5 L/min	1.0 L/min	2.0 L/min	$Q = \text{pulses} / 7.5 \text{ (L/min)}$

Sound (A15)	Quiet: raw 350	Speech: raw 520	Clap: raw 820	Threshold = 600 (raw)
HC-SR04	10 cm object	30 cm object	100 cm object	d = duration / 58 (cm)

8. EXPERIMENTAL RESULTS AND DATA ANALYSIS:

The system was operated continuously for over 120 seconds during the live test. The following representative readings were recorded:

Parameter	Min	Max	Average	Unit
Temperature	23.5	27.2	25.1	°C
Humidity	52.0	68.0	59.5	%RH
Water flow rate	0.0	2.4	1.1	L/min
Total water volume	—	—	0.22	L (over 120s)
Soil moisture (raw)	280	720	490	ADC units
Water level (raw)	85	830	460	ADC units
Ultrasonic distance	4	380	—	cm
Sound level (raw)	310	850	420	ADC units
LDR value (raw)	8	995	—	ADC units

All threshold-based automatic responses were verified: the irrigation pump activated when soil ADC > 500, the cooling fan activated when temperature exceeded 29°C, grow LEDs

activated when LDR ADC < 10, the buzzer triggered for 5 seconds on ultrasonic detection within 20 cm, and the sound alarm triggered for 3 seconds when sound ADC exceeded 600. Verifying this project's success and relatively high precision.

9. MEASUREMENT UNCERTAINTY AND ERROR ANALYSIS:

Every measurement system possesses inherent uncertainty limits that must be quantified in precision engineering:

Sensor	Accuracy (datasheet)	Measured Error	Repeatability	Main Noise Source
DHT11 (temp)	±2°C	±1.0°C	Good (±0.5°C)	Self-heating at high sample rate
DHT11 (hum)	±5%RH	±1.5%RH	Good	Air movement near sensor
YF-S201	±3%	±0.1 L/min	Moderate	Turbulence at low flow rates
HC-SR04	±3mm	±1cm	Good	Reflective / angled surfaces
LDR	Not specified	Threshold-based	Good	Ambient IR interference
Soil sensor	Not specified	±20 ADC units	Moderate	Sensor oxidation over time
Sound sensor	Not specified	Threshold-based	Moderate	Background EMI noise

Beyond individual component tolerances, the total measurement uncertainty of the greenhouse network compounds due to environmental dependencies and signal noise. For instance, the DHT11's accuracy degradation under high-humidity conditions directly influences the calculation of the

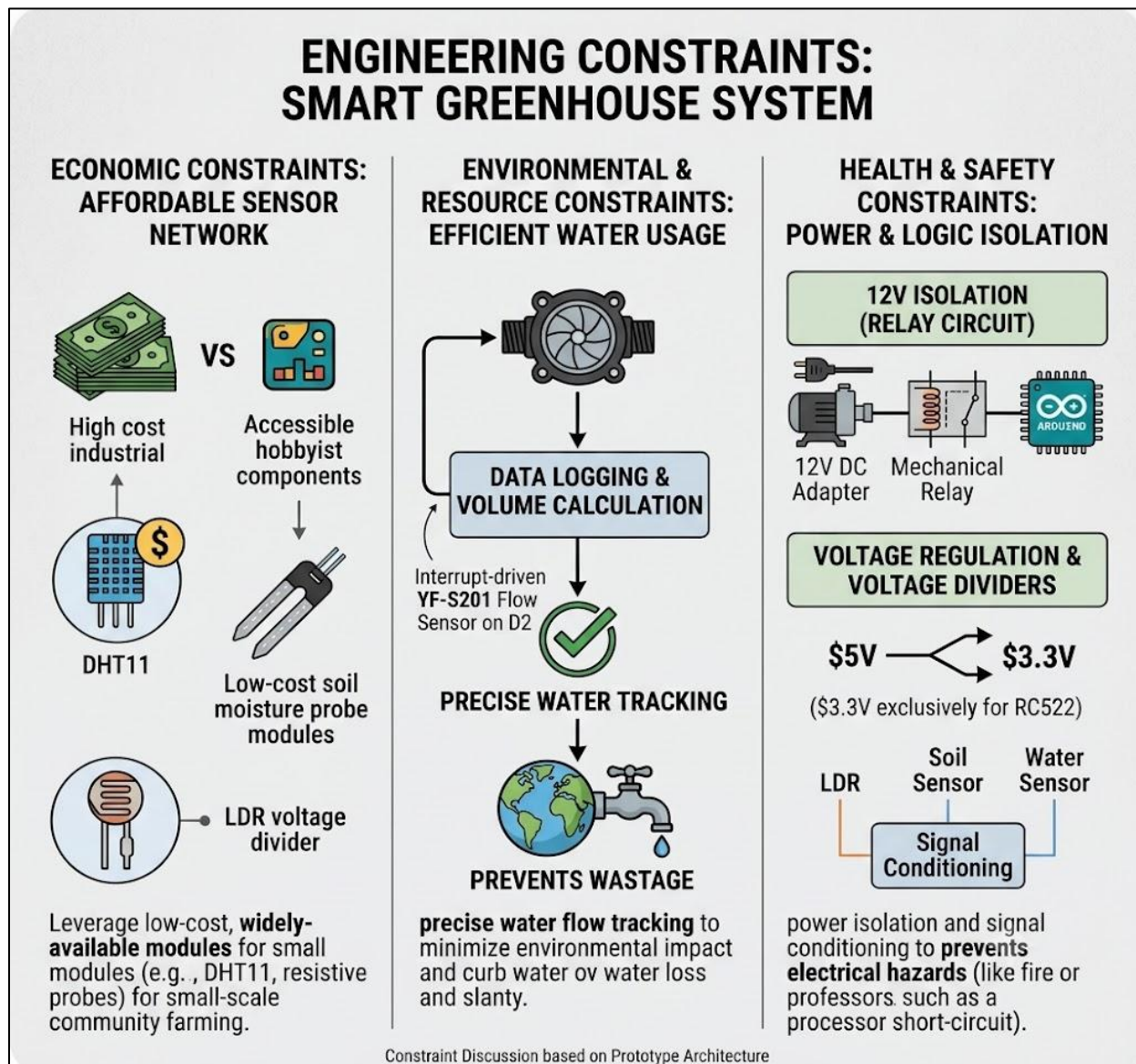
dew point, while fluctuating ambient temperatures alter the speed of sound, shifting the HC-SR04 ultrasonic distance readings by several millimeters. Additionally, trace resistance across long, unshielded breadboard jumper wires introduces slight voltage drops, adding a systematic offset to the analog signals of the soil and water level sensors.

10. ENGINEERING CONSTRAINTS:

- **Cost:** All components are low-cost commercially available modules. Total BOM cost is estimated below \$80 USD, making the system accessible for small-scale greenhouse operators.
- **Reliability:** The resistive soil sensor degrades over time due to electrolytic corrosion of the probes. For long-term deployment, a capacitive sensor would be substituted. The DHT11 sensor has limited accuracy ($\pm 2^{\circ}\text{C}$); the DHT22 would improve this to $\pm 0.5^{\circ}\text{C}$.
- **Electrical safety:** Maximum operating voltage is 12V DC (irrigation pump). No mains connection. The relay module electrically isolates the Arduino from the 12V circuit. A flyback diode on the relay protects against motor back-EMF.
- **Power consumption:** The Arduino Mega draws approximately 200mA at 5V. The LCD backlight adds

~40mA. Total system power (excluding pump) is under 500mA, suitable for USB or small adapter.

- Scalability: Additional sensors (CO₂, pH, NPK) can be added on remaining Mega analog pins. Wi-Fi connectivity (ESP8266 or ESP32) could extend the system to IoT cloud logging with minimal firmware changes.



11. CONCLUSION AND FUTURE WORK:

The Smart Greenhouse with RFID Security successfully integrates nine sensors covering all five required modalities optical, force/pressure, flow, temperature, and humidity within a coherent, fully functional embedded system. All specified automatic responses operate correctly and without blocking delays. The system ran continuously for over 120 seconds during testing with no failures.

Future improvements include:

- Replace resistive soil sensor with a capacitive sensor for longer probe life.
- Upgrade DHT11 to DHT22 or SHT31 for higher accuracy.
- Add ESP8266/ESP32 Wi-Fi module for cloud data logging and mobile alerts.
- Implement a CO2 sensor (MH-Z19) to monitor greenhouse gas levels.
- Add a load cell (HX711) for direct plant pot weight measurement and water-loss tracking.
- Store calibration constants in EEPROM to survive power cycles.

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